

Exam Date & Time: 09-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY 2024

Natural Language Processing [CSE-3172]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What is Natural Language Processing? Discuss with some applications (3)
- A)
- B) What is Morphology? Explain Inflection morphemes with a suitable example. (3)
- C) Draw the Finite state Automaton for the given English verbal Inflection. Explain their Transitions.

Reg-verb-stem	Irreg-verb-stem	Irreg-past-verb	past	Past-part	Pres-part	3sg
walk	cut	caught	-ed	-ed	-ing	-s
fry	speak	ate				
talk	sing	eaten				
impeach	sang					
	spoken					

(4)

- 2) What is CNF? explain how to generate an infinite production rule using CNF? (3)
- A)
- B) Draw finite state transducer (FST) which recognises the suffix -de associated with verbs in English. Demonstrate that your FST will handles cases like 'walk' → 'walked' (5)
- C) What is a regular expression? Write a regular expression to find the words with integers in the text? (2)
- 3) texts = [(3)
- "The price of the product1 is ₹29.99, and the quantity is 100.",
- "The price of the product2 is ₹39.99, and the quantity is 200.",
- "The price of the product1 is ₹49.99, and the quantity is 400."
- A)]

Define a dictionary to store product information
product_info = {'price': [], 'quantity': []}

Write a python code to extract product info like price and quantity from the text and

store them in the dictionary called product_info.

- B) Compute minimum edit distance between BUILDING and BONDING using Levenshtein distance with a cost of 2 for insertions or deletions and 4 for substitutions. (5)
- C) What is Laplace Smoothing? Explain when it is used in n grams model? (2)
- 4) Explain Finite-State Morphological Parsing with a suitable Example? (3)
- A)
- B) For the given sentences:
- Birds can fly in the sky
 - Sky is blue during the day
 - Day turns into night
 - Night is dark and quiet
- (5)
- Learn a bi-gram model using this data. Show the probabilities of all the words and estimate the probability for the sentence "Birds can fly in the sky is blue during the day".
- C) Explain the need of Viterbi algorithm in HMM POS tagging? (2)
- 5) Explain two common kinds of structural ambiguity like attachment ambiguity and coordination ambiguity that occurs during Parsing? (3)
- A)
- B) Consider the given grammar and the string "The bird chased the dog in the garden". Using CKY parsing algorithm parse the string and show the recognition table.
- S → NP VP
 NP → Det N | NP PP
 VP → V NP | VP PP
 PP → P NP
 Det → a | the
 N → dog | bird | garden
 V → chased | caught
 P → in | with
- (5)
- C) Use the following grammar rules create the top-down parse tree for the sentence "the quick brown fox jumped over the lazy dog".
- S → NP VP
 NP → Det Adj Adj N
 VP → V P Det Adj N
 Det → "the"
 Adj → "quick" | "brown" | "lazy"
 N → "fox" | "dog"
 V → "jumped"
 P → "over"
- (2)

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